

Chapter 6

Project Settings

This chapter covers the settings used for defining the properties of each individual project. It's a good idea to familiarize yourself with the information in this chapter prior to setting up your first project.

Contents

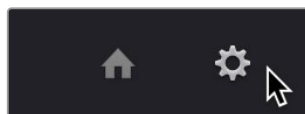
What Are the Project Settings?	138	Broadcast Safe	153
Opening and Editing Project Settings	138	General Options	154
Presets	139	Conform Options	154
Master Settings	141	Color	156
Timeline Format	141	Dynamics Profile	157
Video Monitoring	143	Versions	158
Optimized Media and Render Cache	144	Camera Raw	158
Working Folders	145	Capture and Playback	159
Frame Interpolation	146	Deck Settings	159
Image Scaling	147	Capture	160
Image Scaling	147	Playout	160
Input Scaling	148	Subtitles and Transcription	161
Output Scaling	148	Fusion	161
Color Management	149	Fairlight	162
Color Space and Transforms	149	Timeline Sample Rate	162
Dolby Vision™	150	Bussing	162
HDR10+	151	Audio Metering	162
HDR Vivid	151	Path Mapping	163
Lookup Tables	151	Project Media Locations	164

What Are the Project Settings?

The Project Settings window contains all project-specific parameters that are saved along with that project. These include essential project properties such as the timeline format, video monitoring settings, how to optimize media, and where to save cache files. It also includes image scaling properties, color management settings, and many other properties that affect projects in fundamental ways.

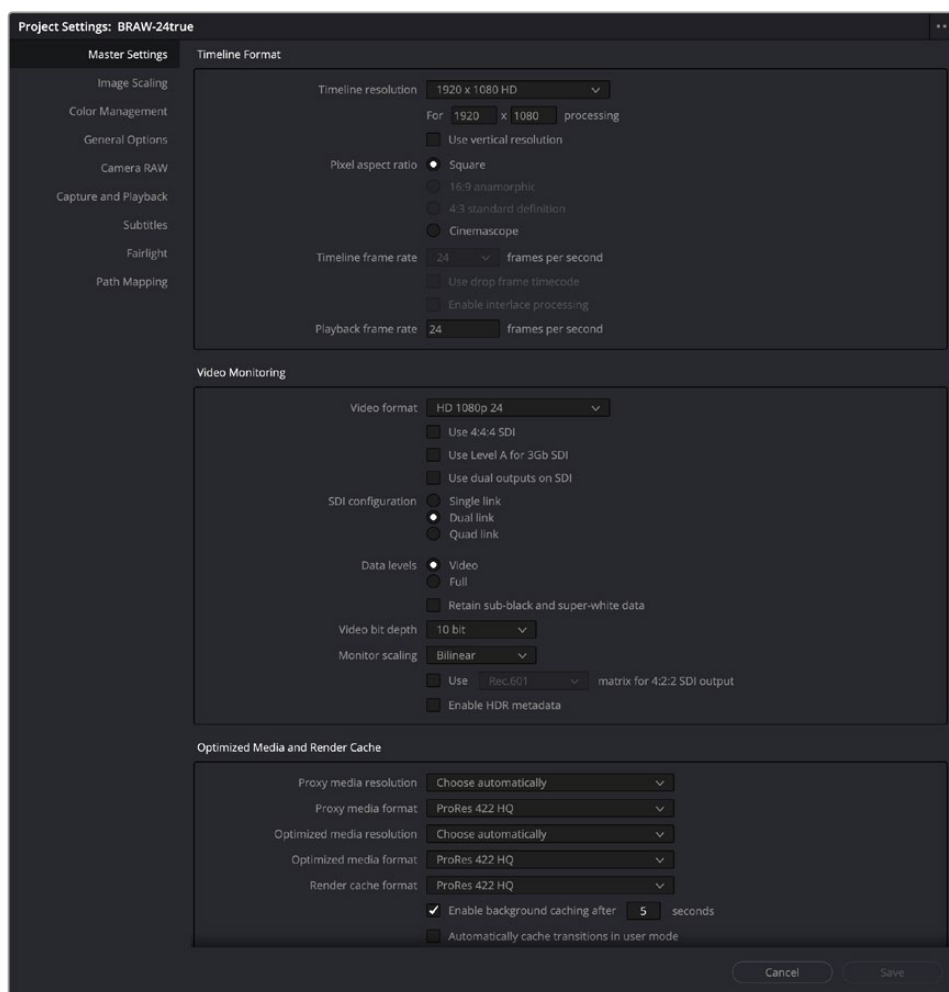
Opening and Editing Project Settings

All of these project-specific settings are easily accessed from anywhere in DaVinci Resolve by clicking the gear button at the bottom right of the page bar.



Project Manager and Project Settings buttons

The Project Settings window opens in the middle of the screen.



Project Settings window

The Project Settings window is divided into a series of panels which can be selected from a sidebar at the left. Each panel contains a collection of related settings that affects some category of DaVinci Resolve functionality.

To alter project settings:

- 1 Click on the name of any group of settings in the sidebar at the left to open that panel.
- 2 Change whatever settings you need to change.
- 3 Do one of the following to apply your changes:

Click Save to apply the changes you've made and close the Project Settings.

Option-click Save to apply the changes you've made and keep the Project Settings window open, so you can make other changes. This option is available because it's sometimes necessary to keep the Project Settings window open as you continue making changes that may visibly affect the clips and timelines in your project.

Presets

The Presets menu lets you save customized collections of Project Settings for future recall. Presets can save the state of nearly every parameter and setting in every panel of the Project Settings window, and make it easy to switch among different setups for different tasks, or to accommodate different types of projects.

The Presets menu is accessed by clicking on the option menu (three dots) in the upper-right corner of the Project Settings window. The Presets menu shows the following items:

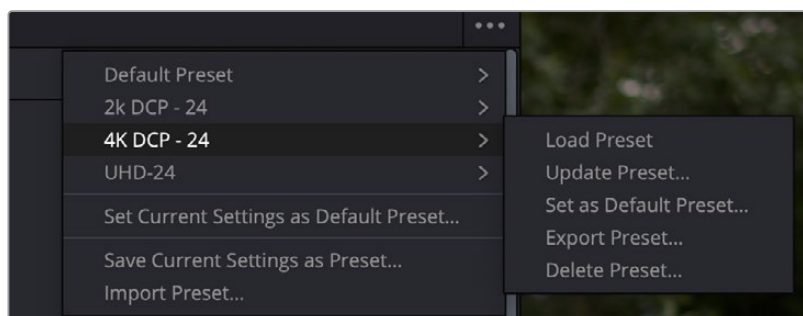
Project Settings Presets: The list of all current Project Presets is shown here, headed by the Default Preset.

Set Current Settings as Default Preset: This option saves the current Project Settings as the new Default Preset.

Save Current Settings as Preset: This option allows you to save the current Project Settings as a new preset that will show up in the menu list above.

Import Preset: This option will open a file browser to let you import a DaVinci Resolve Project Settings Preset file. The extension for these files is .preset.

You can use this menu to create, import, and export your own presets, adding as many as you need to accommodate the types of projects you work on.



The Presets Menu is accessed by clicking on the option menu of the Project Settings window

To create a new preset:

- 1 Do one of the following:
 - Right-click a project in the Project Manager, and choose Project Settings from the contextual menu.
 - Open any project, then open the Project Settings.
- 2 Use the different panels of the Project Settings window to alter whichever settings you need to. There's no need to save your changes as you go; you'll save them all at once later.
- 3 Click on the option menu (three dots) in the upper right-hand corner of the Project Settings window.
- 4 Select Save Current Settings as Preset.
- 5 Enter a descriptive name for your new preset in the dialog box, and press the OK button.
- 6 Your new preset name will now appear in the Project Settings Preset menu.

Once you've created one or more custom presets, you can load them into a project at any time.

To load a preset's settings into a project:

- 1 Open a project with a preset you want to update.
- 2 Open the Project Settings window.
- 3 Click on the option menu in the Project Settings window.
- 4 Click on the Project Preset name you want to use for the open project.
- 5 Select Load Preset from the drop-down menu.

The Project Settings will be updated to the new preset immediately. There is no undo for this function.

To update a preset:

- 1 Open the Project Settings window.
- 2 Use the different panels of the Project Settings window to alter whichever settings you need to update.
- 3 Click on the option menu in the Project Settings window.
- 4 Click on the Project Preset name you want to update.
- 5 Select Update Preset from the drop-down menu, and select Update from the dialog box.

The preset will be replaced with your current settings. There is no undo for this function.

To set the default preset:

- 1 Select the preset from the Project Settings Presets list you want to be loaded each time you make a new project.
 - 2 Click on the preset and select Set As Default Preset.
- or
- 1 Select "Set Current Settings as Default Preset" from the option menu of the Project Settings window to make whatever your project settings are currently the new default preset.
 - 2 Select Update from the dialog box.

The preset will be replaced with your current settings. There is no undo for this function.

To export a preset:

- 1 Click on the preset from the Project Settings Presets list you want to export.
- 2 Choose Export Preset from the drop-down menu.
- 3 Enter a name for the preset and where you want to save it in the file browser, and press Save.

The DaVinci Resolve Project Settings Preset file will export to that destination with a “.preset” extension.

To import a preset:

- 1 Click on the option menu in the Project Settings window.
- 2 Select Import Preset.
- 3 Navigate the file browser to the DaVinci Resolve Project Settings Preset file you wish to import. The file will have a “.preset” extension. Press Open.

To delete a preset:

- 1 Click on the preset from the Project Settings Presets list you want to delete.
- 2 Choose Delete Preset from the drop-down menu.
- 3 Press Delete in the dialog box.

The Project Settings will be deleted immediately. There is no undo for this function.

Master Settings

This panel is project specific and lets you set up and adjust the most essential properties of the timelines in your project, including the timeline format, video monitoring method, and conform options. In many workflows, you'll want to adjust these settings before getting started with your project.

By default, all timelines use these project-wide settings. However, beginning with DaVinci Resolve 16, you can optionally create timelines with individual Format, Monitoring, and Output Sizing settings. However, if you change a timeline to use “Basic Settings,” then that timeline will mirror the project-wide options that are selected in the Project Settings.

Timeline Format

This group of settings affects the geometry and image processing of the current project.

- **Timeline resolution:** A drop-down menu that lets you choose a frame resolution preset to use for image processing while grading. DaVinci Resolve is resolution independent, so you can change the resolution at any time and all windows, tracks, sizing changes, and keyframe data will be automatically recalculated to fit the new size. For example, you can work on a 4K project while monitoring at HD resolutions if your room is only set up with an HD monitor, and then render the finished project at 4K resolution for final delivery. Alternately, you can downsize an HD project to an SD resolution to create another set of deliverables.

For more information on Resolve's resolution independence, see *Chapter 152, “Sizing and Image Stabilization.”*

- **Frame size (Labeled “For X x Y processing”):** Lets you set resolutions not found in the “Timeline resolution” drop-down menu.

- **Use vertical resolution:** This checkbox swaps the horizontal and vertical pixels of the Timeline resolution. This lets you format your timeline vertically for display on smart phones, tablets, or televisions that are in an upright configuration for digital signage.
- **Pixel aspect ratio:** Used to select PAR settings for image formats that don't use the default square pixel format. You can apply a 16:9 anamorphic PAR, a 4:3 PAR for SD projects, or a Cinemascope ratio.
- **Timeline frame rate:** Determines the primary frame rate used by the project. A variety of standard and high frame rate (HFR) settings are available. If you're importing an AAF or XML file, this setting is automatically set via an option in the Project Import dialog. Ideally, you should choose a frame rate before importing media into the Media Pool. However, the first time you import media into an empty Media Pool, you're prompted if the incoming media frame rate doesn't match the Timeline frame rate set here, and you have the option of automatically updating this setting to match that of the media you're importing. Once one or more files have been added to the Media Pool, this setting cannot be changed.
- **Use drop frame timecode:** Enables or disables drop frame timecode for the current project. Off by default.
- **Enable interlace processing:** Interlaced media is supported throughout DaVinci Resolve. The "Enable interlace processing" checkbox forces DaVinci Resolve to process all operations internally using separated fields, in order to properly maintain the field integrity of interlaced clips in your program. In addition, each clip in the Media Pool has a Field Dominance drop-down menu in the Video panel of the Clip Attributes window that lets you specify whether clips are upper- or lower-field dominant; an Auto setting makes this choice by default.

There is also a corresponding checkbox in the Render Settings panel of the Deliver page, named "Field rendering," that lets you enable and disable field rendering when you're rendering file-based output.

There are two instances where you want to leave this setting turned off:

- If you're working with progressive-frame media, it is not necessary to turn this checkbox on. Doing so will unnecessarily increase processing time.
- If you're using interlaced clips in a progressive-frame project and you're intending to deinterlace those clips using the Enable Deinterlacing checkbox in the Clip Attributes window, then you must keep "Enable video field processing" off. Otherwise, the Enable Deinterlacing checkbox will be disabled for all clips.

For more information about deinterlacing clips, see *Chapter 22, "Modifying Clips and Clip Attributes."*

If you're working on a project with interlaced media that you intend to keep interlaced, then whether or not it's necessary to turn field processing on depends on what types of corrections you're applying to your clips. If you're mastering your program to an interlaced format, and you're applying any adjustments that would cause pixels from one field to move or bleed into adjacent fields, then field processing should be enabled; effects requiring field processing include filtering operations such as blur, sharpen, and OpenFX operations, as well as sizing transforms that include pan, tilt, zoom, rotate, pitch, and yaw.

On the other hand, regardless of whether you're outputting interlaced or progressive-frame media, if you're not filtering or resizing your clips, and you're only applying adjustments to color and contrast, it's not necessary to turn on field processing for interlaced material, and in fact, leaving it off may somewhat shorten your project's rendering time.

- **Align Clips to Frame Boundaries:** When working with interlaced material, you can choose whether to perform edits at the field or frame boundaries. When checked this means all editing actions and playhead stepping will occur at full frame boundaries. This works independently of the audio subframe controls, so it's still possible to get perfect audio sync while working with video at the frame level. When unchecked, all editing actions and playhead stepping will be performed at the field boundary instead.
- **Playback frame rate:** Usually mirrors the frame rate selected in the "Video format" setting (in the Video Monitoring section below), which is typically based on the frame rate of the external display that's connected to your video interface, given the "Timeline Frame Rate" setting. For example, a 50Hz monitor requires a 25 fps playback frame rate for synchronous display without dropped frames. If you want to monitor playback at a slower frame rate, type the frame rate of your choice in this field and DaVinci Resolve will make the appropriate calculations to drop or repeat frames as necessary to match it. This can be useful for temporarily seeing how clips look in slow motion.

Video Monitoring

The settings available in this group control the signal that's output by the video output interface that's connected to your workstation, and let you specify what standard of signal is output, and via which signal path.

By default the frame size and frame rate match those in the Timeline resolution and Playback frame rate options. However, if necessary you can change these settings to match those of the external display you're using to monitor your work. For example, if you're working with 2K files for 2K output, but you're color correcting using a high definition monitor set to 1080 resolution, you can select the appropriate HD standard for that monitor without changing the Timeline Resolution settings.

- **Video Resolution:** Lets you choose a standard video frame size to be output via your connected video output interface.
- **Format:** Lets you choose a standard video frame rate to be output via your connected video output interface.
- **Video connection checkboxes:** Lets you choose the signal standard to output from your connected video output interface to the video monitor. Make sure to choose a standard that's supported by both your video interface and your monitor.

The options are:

Use 1080PsF: Output progressive video in 1080i format for older HDTV sets.

Use 4:4:4 SDI: A signal path for monitoring image data to monitors that supports 4:4:4 chroma sampling, typically over SDI connections.

Use Level A for 3Gb SDI output: A signal path for monitoring image data via a single 3 Gb/s SDI connection.

Use dual outputs on SDI output: All DaVinci Resolve systems can generate a side-by-side display that can be sent to a Stereoscopic monitor via the HD-SDI output of an UltraStudio 4K or DeckLink card. When dual SDI 3D monitoring is enabled, each eye is output separately at full resolution. In this mode, split-screen wipes and cursors will not be visible on the grading monitor.

- **SDI Configuration:** Lets you choose from among Single Link, Dual Link, and Quad Link SDI, depending on what your display supports.
- **Data Levels:** This setting only affects the data levels being output via the video interface that connects the DaVinci Resolve workstation to your external display. It has no effect on the data that's processed internally by DaVinci Resolve, or on the files written when you render in the Deliver page. It is imperative that the option you choose in DaVinci Resolve matches the data range to which your external display is set. Otherwise, the video signal will appear to be incorrect, even though the internal data is being processed accurately by DaVinci Resolve.

There are two options:

Video: This is the correct option to use when using a broadcast display set to the Rec. 709 video standard.

Full: If your monitor or projector is capable of displaying “full range” video signals, and you wish to monitor the full 10-bit data range (0–1023) while you work, this is the correct option to use.

For more information about data levels, see *Chapter 9, “Data Levels, Color Management, and ACES.”*

- **Retain sub-black and super-white data:** Turning this checkbox on lets DaVinci Resolve output the undershoots (sub-black) and overshoots (super-white) within the headroom of video encoded data levels to video. When this is turned off, these out-of-bounds values are clipped in video output.
- **Video bit depth:** Choose the bit depth that corresponds to the capability of your display. You can choose between 8-bit and 10-bit. Monitoring in 10-bit is more processor intensive, but preferable to avoid the appearance of banding that may not in fact be in the image data being processed by DaVinci Resolve.
- **Monitor scaling:** Defaults to basic, and is only enabled to smooth the edges of video being viewed on a projector with very large screens. These settings minimize high frequency artifacts that may be seen. This may also be noticeable if you have a 2K or HD project but are monitoring on an SD monitor. The other option, Bilinear, has different effects on the monitored image depending on your display device, so you may need to check to verify that it’s appropriate for your environment.
- **Use Rec601 Matrix for 4:2:2 SDI output:** Don’t use this checkbox unless you know what it does. You know who you are.
- **Enable HDR metadata:** (only available in Studio version) Turning on this checkbox outputs the metadata necessary to send High Dynamic Range signals over HDMI 2.0a and have it be correctly decoded by an HDR-aware video display. When this checkbox is enabled, it’s recommended to also enable the “HDR mastering is for X nits” checkbox in the Color Management page, and set the “nit” level (slang for cd/m2) to whatever peak luminance level your HDMI connected HDR display is capable of.

Optimized Media and Render Cache

These settings govern the resolution and codec of optimized media that DaVinci Resolve can generate in order to facilitate greater real time performance, as well as cached media that’s generated by the Smart and User Cache.

- **Proxy media resolution:** A drop-down list lets you choose whether to generate proxy media at each clip’s Original size, or at Half, Quarter, One-Eighth, or One-Sixteenth the resolution of the original media, or allow DaVinci Resolve to choose this automatically for you based on your timeline settings.
- **Proxy media format:** Specifies the format in which proxy media files will be written. You can choose from among a variety of Uncompressed, ProRes, and DNxHD formats, depending on your requirements.
- **Optimized media resolution:** A drop-down list lets you choose whether to generate optimized media at each clip’s Original size, or at Half, Quarter, One-Eighth, or One-Sixteenth the resolution of the original media, or allow DaVinci Resolve to choose this automatically for you based on your timeline settings.
- **Optimized media format:** Specifies the format in which optimized media files will be written. You can choose from among a variety of Uncompressed, ProRes, and DNxHD formats, depending on your requirements.

- **Render cache format:** Specifies the format in which render cache files will be written. You can choose from among a variety of Uncompressed, ProRes, and DNxHD formats, depending on your requirements.
- **Enable background caching after X seconds:** Specifies the duration of inactivity after which automatic background caching will begin.

A series of checkboxes let you force specific types of effects to be cached when you use the User Cache, which is a more selective manner of caching than the Smart Cache. These include:

Automatically cache transitions in user mode: If you're using User mode and you find that your workstation does not have adequate performance to play transition effects in real time, you can force these categories of effects to be automatically included in the Sequence Cache and cached when you're using the User mode of caching.

Automatically cache composites in user mode: If you're using User mode and you find that your workstation does not have adequate performance to play composite mode or opacity effects in real time, you can force these categories of effects to be automatically included in the Sequence Cache and cached when you're using the User mode of caching.

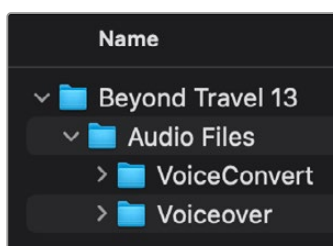
Automatically cache Fusion effects in user mode: If you've created effects for a clip in the Fusion page and you find that your workstation does not have adequate performance to play that clip in real time, you can force these categories of effects to be automatically included in the Sequence Cache and cached when you're using the User mode of caching.

Working Folders

These fields let you specify to which folders cache and gallery files are written.

- **Project media location:** Lets you choose a specific folder on your system to hold all generated media created for that project. Generated media includes things like new audio files created from the Voiceover tool and Voice Convert, or essentially any type of new media created by DaVinci Resolve itself. This allows you to make sure all the media for a project is in one common location, making it easier to hand off, move or relink media, rather than have some media in DaVinci Resolve's internal file structure and some on a media drive. DaVinci Resolve will automatically create new subfolders at your chosen media location based on media type, which makes it easy to send some or all generated media to another location.

You can also set this location in the Create New Project dialog when creating a new project.



Inside the project media location, DaVinci Resolve will create subfolders automatically to organize any media generated from within the program.

- **Proxy generation location:** All proxy media files that you create are saved in the directory path specified by this field.
- **Cache files location:** All render cache files that you create are saved in the directory path specified by this field. This path defaults to a hidden "CacheClip" directory that's created at the location of the first Media Storage Volume you specify in the DaVinci Resolve Preferences window.

- **Gallery stills location:** By default, all stills you save are saved in the DPX format, and are placed in the directory path specified by this field. This path defaults to a hidden “.gallery” directory that’s created at the location of the first Media Storage Volume you specify in the DaVinci Resolve Preferences window.

NOTE: If the volume you’ve selected to use for the cache becomes unavailable, DaVinci Resolve will warn you with a dialog.

Frame Interpolation

These settings determine the default state for all retiming and speed change effects, including when clips are in mixed frame rate timelines.

- **Retime Process:** This drop-down menu lets you choose a default method of processing clips that don’t match the project frame rate in mixed frame rate timelines and clips with speed effects (fast forward or slow motion) applied to them, throughout the project. Since each clip in every timeline defaults to “Project Settings,” changing this setting will change the way most mixed frame rate and speed effected clips will be processed, except for those with custom settings selected.

There are three options:

Nearest: The most processor efficient and least sophisticated method of processing; frames are either dropped for fast motion, or duplicated for slow motion.

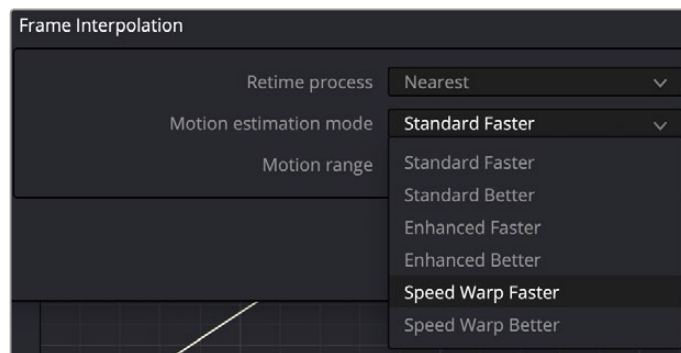
Frame Blend: Also processor efficient, but can produce smoother results; adjacent duplicated frames are dissolved together to smooth out slow or fast motion effects. This option can provide better results when Optical Flow displays unwanted artifacts.

Optical Flow: The most processor intensive, but highest quality method of speed effect processing. Using motion estimation, new frames are generated from the original source frames to create slow or fast motion effects. The result can be exceptionally smooth when motion in a clip is linear. However, two moving elements crossing in different directions or unpredictable camera movement can cause unwanted artifacts.

- **Motion estimation mode:** When using mixed frame rate clips in a timeline that has Optical Flow retiming enabled, when using Optical Flow to process speed change effects, or when using Image Stabilization or Temporal Noise Reduction controls in the Color page, the Motion Estimation drop-down of the Master Settings (in the Project Settings window) lets you choose options that control the trade-off between speed and quality.

There are additional “Enhanced” Optical Flow settings available in the “Motion estimation mode” drop-down in the Master Settings panel of the Project Settings. The “Standard Faster” and “Standard Better” settings are the same options that have been available in previous versions of DaVinci Resolve. They’re more processor-efficient and yield good quality that are suitable for most situations. However, “Enhanced Faster” and “Enhanced Better” should yield superior results in nearly every case where the standard options exhibit artifacts, at the expense of being more computationally intensive, and thus slower on most systems.

“Speed Warp Faster” and “Speed Warp Better” are available for even higher-quality slow motion effects using the DaVinci Neural Engine. Your results with this setting will vary according to the content of the clip, but in ideal circumstances this will yield higher visual quality with fewer artifacts than even the Enhanced Better setting.



New improved Motion estimation mode settings in the Master Settings panel of the Project Settings

- **Motion range:** When using mixed frame rate clips in a timeline that has Optical Flow retiming selected, or when using Optical Flow to process speed change effects, this drop-down menu lets you choose the default setting to use, small, medium or large motion, for all speed and motion related calculations so you can try and improve the result by matching the type of motion in the source media. This setting can also be changed on a clip by clip basis in the Edit page Inspector.

Image Scaling

The Image Scaling panel contains settings that determine how and when clips are resized for various reasons.

Image Scaling

These settings affect the methods used to resize clips in various situations.

- **Resize Filter:** The first group of settings lets you choose the filter method used to interpolate image pixels when resizing clips:

Smoother: May provide higher quality for projects using clips that must be scaled down to fit an SD resolution frame size.

Bicubic: While the Sharper and Smoother options are slightly higher quality, Bicubic is still an exceptionally good resizing filter and is less processor intensive than either of those options.

Bilinear: A lower quality setting that is less processor intensive. Useful for previewing your work on a low-performance computer before rendering, when you can switch to one of the higher quality options.

Sharper: Usually provides the best quality in projects using clips that must be scaled up to fill a larger frame size or scaled down to HD resolutions.

Custom: This setting lets you take control of the exact algorithm used in all resizing operations. The custom Resize Filter options available are: Bessel, Box, Catmul-Rom, Cubic, Gaussian, Lanczos, Mitchell, Nearest Neighbor, Quadratic, and Sinc. In practice, the difference between these methods can be quite subjective. However, if you need to match a specific resizing method used from another application, you can do it here. For everyday use, the normal resizing filters in DaVinci Resolve should be sufficient.

- **Override input scaling:** Checking this box lets you choose an Input Sizing preset to apply to the project.
- **Override output scaling:** Checking this box lets you choose an Output Sizing preset to apply to the project.

- **Anti-alias edges:** A second group of settings lets you choose how to handle edge anti-aliasing for source blanking.

Auto: Adds anti-aliasing when any of the Sizing controls are used to transform the image. Otherwise, anti-aliasing is disabled.

On: Forces anti-aliasing on at all times.

Off: Disables anti-aliasing. It might be necessary to turn anti-aliasing off if you notice black blurring at the edges of blanking being applied to an image.

- **Deinterlace quality:** (only available in Studio version) A fourth group of settings lets you choose the quality/processing time tradeoff when deinterlacing Media Pool clips using the Enable Deinterlacing checkbox in the Clip Attributes window.

There are three settings:

Normal: A high-quality deinterlacing method that is suitable for most clips. For many clips, Normal is indistinguishable from High. Normal is always used automatically during playback in Resolve.

High: A more processor-intensive method that can sometimes yield better results, depending on the footage, at the expense of slower rendering times.

DaVinci Neural Engine: This option uses the advanced machine learning algorithms of the DaVinci Neural Engine to analyze motion between the fields of interlaced material and reconstructs them into a single frame. This option is very computationally intensive but, ideally, will deliver an even more aesthetically pleasing result than the “high” setting.

Input Scaling

Contains one setting, Mismatched resolution files, that lets you choose how clips that don't match the current project resolution are handled. There are four options:

- **Center crop with no resizing:** Clips of differing resolution are not scaled at all. Clips that are smaller than the current frame size are surrounded by blanking, and clips that are larger than the current frame size are cropped.
- **Scale full frame with crop:** Clips of differing resolution are scaled so that the clip's shortest dimension is fit to match the frame. Excess pixels are cropped.
- **Scale entire image to fit:** The default setting. Clips of differing resolution are scaled so that the clip's longest dimension is fit to match the frame. The shorter dimension has blanking inserted (letterboxing or pillarboxing).
- **Stretch frame to all corners:** Useful for projects using anamorphic media. Clips of differing resolutions are squished or stretched to match the frame size in all dimensions. This way, anamorphic media can be stretched to match full raster, or full raster media can be squished to fit into an anamorphic frame. An added benefit of this setting is that it makes it easy to mix anamorphic and non-anamorphic clips in the same project.

Output Scaling

These settings let you optionally choose a different resolution to be output via your video output interface, for monitoring, outputting to tape, or rendering. In particular, if you set the resolution in the Render Settings list of the Deliver page to something other than the Timeline Resolution, these settings are used to make the change (for example, if you're rendering a downconversion of the current timeline). This can be used in situations where you're working on a high resolution 4K project, but you want to monitor using an HD display and output HD resolution media for approval.

- **Match timeline settings:** Turned on by default, so that these settings mirror the Timeline Resolution, Image Scaling, and Input Image Scaling settings described above. Turning this checkbox off lets you choose different settings for monitoring, outputting to tape, or rendering, using the other settings in this group.

- **Output resolution:** Lets you choose an alternate resolution.
- **For:** Lets you specify a different custom alternate resolution.
- **Pixel aspect ratio:** Lets you specify an alternate pixel aspect ratio to match the alternate timeline format.
- **Mismatched resolution files:** Lets you choose an alternate way of handling mismatched resolution files given the alternate resolution you've chosen. These options work identically to those of the "Input Image Scaling" group.
- **Super Scale:** Sets a very processor-intensive and high quality upscaling algorithm that actually creates new pixels for the resized image. The possible values are: None, 2-3-4x, 2-3-4x Enhanced, and Auto. For more information on Super Scale, see Chapter 11, "Image Sizing and Resolution Independence."

For more information on Super Scale, see *Chapter 11, "Image Sizing and Resolution Independence."*

Color Management

The various options found in the Color Management panel let you configure DaVinci Color Management (RCM) or ACES if you have either enabled, and they also allow you to pre- or post-process the DaVinci Resolve image processing pipeline using LUTs and Broadcast Safe settings, in order to accommodate a wide range of different color workflows.

Color Space and Transforms

If you choose DaVinci YRGB Color Managed or ACES in the Color Science menu at the top, then the other drop-down menus in this section become enabled.

For more information about DaVinci Resolve Color Management and ACES, see *Chapter 9, "Data Levels, Color Management, and ACES."* If you're new to color or color management, you're strongly recommended to read this chapter.

If you choose to use Resolve Color Management (RCM), ACEScc, or ACEScct, the settings in this panel give you extensive control over how color is transformed, starting with choosing the default color settings for the source media in your project (via the Input Color Space), through choosing how you want your grading controls in DaVinci Resolve to behave (via the Timeline Color Space), and then specifying how the final color will look on your monitor and output device (via the Output Color Space).

- **Color science:** There are four options that let you choose whether to work with manual or automated color management.
 - DaVinci YRGB color science:** DaVinci Resolve's original color science, in which you manage all and any color transforms from one color space to another manually, using either LUTs or manual adjustments.
 - DaVinci YRGB Color Managed:** Enables the Resolve color-managed workflow (RCM) for grading.
 - DaVinci ACEScc or ACEScct:** Both of these are standardized color management schemes that are available for facilities using ACES workflows. Of the available settings, ACEScct is the most intuitive way of working for most colorists, as it handles the lifting of shadows in a creatively useful way.

For more information about Color Management and ACES, see *Chapter 9, “Data Levels, Color Management, and ACES.”*

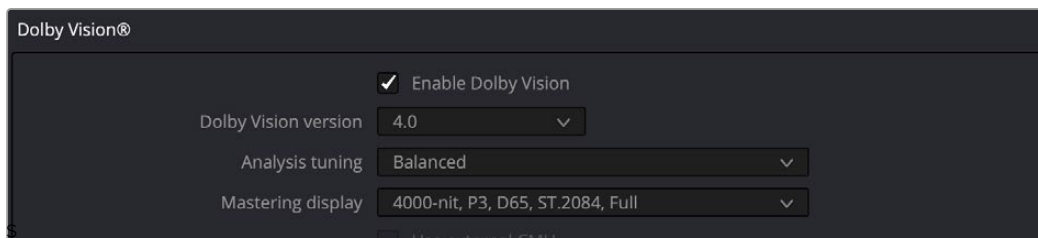
- **ACES version:** This drop-down only appears if you choose one of the DaVinci ACES options from the Color science drop-down menu. Lets you switch between different versions of the ACES specification. This lets you choose the appropriate older version of ACES whenever you open an older project.
- **Use Separate Color Space and Gamma:** If this checkbox is turned off (the default), the Color Management panel of the Project Settings exposes one drop-down each for the Input, Timeline, and Output Color Space settings, and each setting simultaneously transforms the gamut and gamma, depending on which option you choose. If you turn this checkbox on, then the Color Management panel changes so that the Input, Timeline, and Output Color Space settings each display two pop-ups. The first drop-down lets you explicitly set the gamut, while the second drop-down lets you explicitly set the gamma.

To provide more detailed information, the simple and advanced global controls available for Resolve Color Management (RCM) are covered in a dedicated chapter.

For more information, see *Chapter 9, “Data Levels, Color Management, and ACES.”*

Dolby Vision™

DaVinci Resolve includes a GPU-accelerated software version of the Dolby Vision CMU (Content Mapping Unit) for doing Dolby Vision grading and finishing workflows right in either the free version of DaVinci Resolve or in DaVinci Resolve Studio. This is enabled and set up in the Color Management panel of the Project Settings with the Enable Dolby Vision checkbox.



Dolby Vision settings in the Color Management panel of the Project Settings

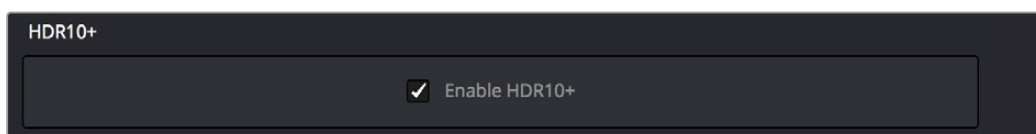
There are five controls available:

- **Enable Dolby Vision:** Turns Dolby Vision on and off. When on, this checkbox enables the Dolby Vision palette in the Color page.
- **Dolby Vision version drop-down:** Lets you choose which version of the Dolby Vision algorithms to use. Options at the time of this writing include 2.9 and 4.0.
- **Analysis tuning:** Choose from a variety of options for how Dolby Vision will analyze and control highlight retention.
- **Master Display drop-down:** Lets you choose the nit level and gamut of the master HDR display you're grading on.
- **Use External CMU:** A checkbox lets you choose whether to use the built-in software CMU or a hardware CMU that you have connected to your DaVinci Resolve workstation.

NOTE: Dolby Vision controls are available to all DaVinci Resolve users for monitoring and automatically generating Dolby Vision metadata for creating other HDR and SDR deliverables from the HDR grade you've made. However, if you want to be able to make manual trims on top of this automatic analysis, you must email dolbyvisionmastering@dolby.com for more information on obtaining a license.

HDR10+

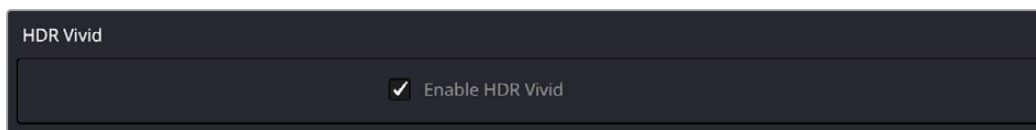
DaVinci Resolve supports the new HDR10+ HDR format by Samsung. Please note that this support is a work in progress as this is a new standard. When enabled, an HDR10+ palette exposes trimming parameters that let you trim an automated downconversion of HDR to SDR, creating metadata to control how HDR-strength highlights look on a variety of supported televisions and displays. This is enabled and set up in the Color Management panel of the Project Settings with the Enable HDR10+ checkbox. Turning HDR10+ on enables the Dolby Vision palette in the Color page.



HDR10+ settings in the Color Management panel of the Project Settings

HDR Vivid

HDR Vivid is the HDR video technology standard released by the China UHD Video Industry Alliance (CUVA). Mastering in this format assures wide compatibility with HDR televisions, phones, computers, and other devices in China.

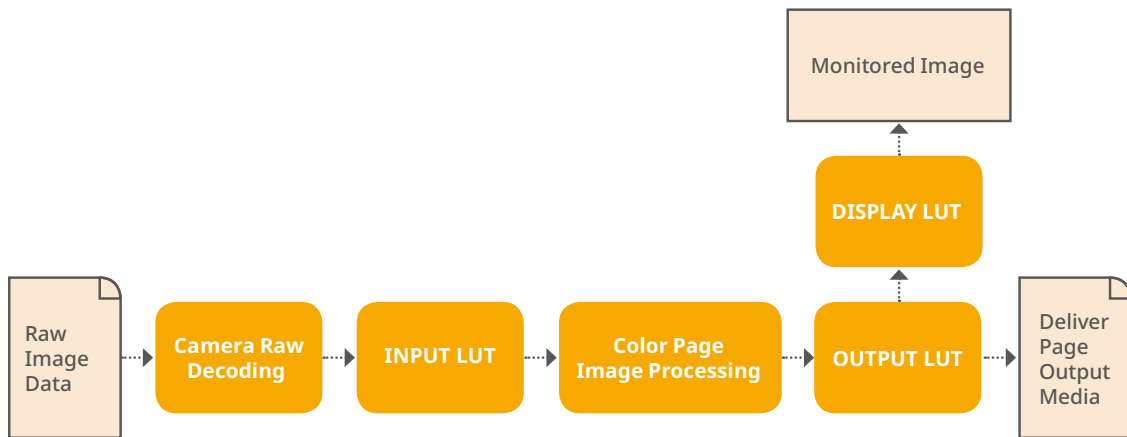


HDR Vivid settings in the Color Management panel of the Project Settings

Lookup Tables

This group of controls lets you add LUTs to the Resolve image processing pipeline that affect every timeline in the entire project all at once. These LUTs can be used for a wide variety of functions, such as to trim Timeline grades, apply Log to Linear conversions, simulate film output, and limit the signal to accommodate Broadcast Safe requirements.

Different options let you insert image processing to different stages of the pipeline as seen in the following diagram:



Keep in mind that since you can apply both 1D and 3D LUTs simultaneously, 1D LUTs at each step are always applied before 3D LUTs.

- **Input Lookup Table:** Two drop-down menus let you add 1D and/or 3D LUTs that process the current Timeline before every other image processing operation in DaVinci Resolve.
- **Output Lookup Table:** Two drop-down menus let you add 1D and/or 3D LUTs that process the current Timeline after the operations applied in the Color page, but before the temporarily applied Display LUT.
- **Video Monitor Lookup Table:** Two drop-down menus let you add 1D and/or 3D LUTs that process the current Timeline after every other image processing operation in DaVinci Resolve. However, Display LUTs are only temporarily applied for purposes of monitoring; they're never applied to rendered media, or to the signal that is output to tape using the controls in the Deliver page. Display LUTs are particularly valuable for applying a film print emulation LUT in a Log workflow, or for applying a monitor calibration LUT if you're outputting to a single display and you don't have dedicated outboard calibration hardware.

Here's an example. It's common, when grading for film output using a Log workflow, that you'll use the Display LUT drop-down menu to apply a film emulation LUT that simulates the image as it will be output from the film recorder, taking into account the film lab and print stock used, in order to make sure that the image you're grading will appear as close as possible to what the eventual release print will look like in the cinema.

- **Color Viewer Lookup Table:** Two drop-down menus let you add 1D and/or 3D LUTs that process the image shown in the Viewer on your computer display, independently of the Display LUT that's used to output to your broadcast display. By default, this follows the Video Monitor LUT setting, but you can also use this option to apply a specific calibration transform for your computer monitor. Alternately, you could use it to desaturate the GUI Viewer to be able to specifically evaluate image contrast, or if you don't want to have to argue with your client over which display looks correct.
- **Scopes Lookup Table:** Ordinarily, DaVinci Resolve's internal software video scopes provide an unbiased analysis of the actual video data levels within the Resolve image processing pipeline. However, you can choose to have the software scopes use the Video Monitor LUT selection, or any other LUT installed on your system, to transform this analysis to reflect the monitored output.
- **3D Lookup Table Interpolation:** Lets you choose the processing quality of both LUT and DCTL operations in DaVinci Resolve. 3D Lookup tables (LUTs) are 3D tables of red, green, and blue values that specify an output color value for each input color value, thereby providing a method of making color transformations using pre-calculated data. While powerful, 3D LUTs have finite detail; for example, one might have a 17x17x17 LUT that specifies 4913 individual color transforms. When applied to a floating point image that contains more data than the LUT specifies transforms

for, color values falling between the 17x17x17 color transforms specified by the LUT need to be interpolated. You can choose from two methods that trade off processing efficiency for higher quality:

Trilinear: (Default) Trilinear is backward compatible with grades that use LUTs from previous versions of DaVinci Resolve and matches the look of LUTs being applied in other applications.

Tetrahedral: Tetrahedral is slightly more processor-intensive, but results in higher image quality LUT and DCTL processing, with reduced color-banding. Tetrahedral is recommended for projects that don't need backward compatibility with previous versions of DaVinci Resolve or LUTs created in other applications.

- **Update Lists button:** Refreshes the LUT drop-down menus if you've added new LUTs to your system since DaVinci Resolve has been opened.
- **Open LUT Folder button:** This selection opens the master folder in your file system, as described in the list of DaVinci Resolve LUT paths shown above.

Adding Lookup Tables to Your DaVinci Resolve Installation

The drop-down menus in the Color Management panel include a series of factory preset LUTs that were installed with DaVinci Resolve, along with any LUTs that have been generated by DaVinci Resolve, or that you've imported into the proper directory for your operating system.

- **On macOS:** Library/Application Support/Blackmagic Design/DaVinci Resolve/LUT/
- **On Windows:** C:\ProgramData\Blackmagic Design\DaVinci Resolve\Support\LUT
- **On Linux:** /opt/resolve/LUT

If you downloaded the non-studio version of DaVinci Resolve from the Apple App Store, LUTs are saved in a different location in order for DaVinci Resolve to remain totally self-contained. In this case, you can click the "Open LUT Folder" button in the Lookup Tables panel of the Project Settings, to open up a Finder window at the location these LUTs are stored. You can use this window to copy LUTs that you want Resolve to have access to, or delete LUTs that you no longer need.

If you add a LUT to one of these directories after DaVinci Resolve has been opened, you can click the Update Lists button to refresh the contents of the drop-down menus.

DaVinci Resolve uses both 1D and 3D LUTs. 3D LUTs that are created by DaVinci Resolve are in the .cube format, configured as 33x33x33 cubes with 32-bit floating point processing. DaVinci Resolve can also read and use LUTs in the Shaperlut format.

Broadcast Safe

Broadcast Safe settings can be enabled while you grade to limit both the luma and chroma of the video signal to one of three levels of acceptable overshoots and undershoots.

- **Broadcast safe IRE (mV) levels:** A drop-down menu for choosing one of three levels of aggressiveness when limiting the signal. Choose the range that corresponds to your QC requirements.
- **Make Broadcast Safe:** A checkbox that turns broadcast safe limiting on and off.

NOTE: The clipping imposed by Broadcast Safe itself does not have an inherently soft roll-off. For best results, Broadcast Safe should be used in conjunction with the Soft Clip controls in the Color page.

General Options

This panel presents a selection of general preferences that affect the interface and operation of DaVinci Resolve.

Conform Options

The settings in this group determine how clips are conformed to match imported project files with source media on disk by extracting timecode, reel names, file names, file paths, and so on.

For more information on conforming and relinking, see *Chapter 56, “Conforming and Relinking Clips.”*

- **Use Timecode:** Determines how DaVinci Resolve extracts timecode from referenced media files. There are two options:

Embedded in the source clip: The preferred setting for most projects to conform automatically and apply grades to the resulting clips. As long as DaVinci Resolve can reference the timecode in either a media file’s timecode track, or in the header metadata of the frames in a DPX sequence, you can use timecode to reconform clips, or even completely change the media file to which a clip refers.

From the source clip frame count: This setting is useful if the source media lacks timecode metadata, and all that’s available is a frame count that identifies frames via sequentially numbered integer values.

- **Conform partial clips with black gaps:** Inserts black frames whenever you conform a clip that doesn’t contain all the required frames. When this option is selected, partial clips are flagged in the Edit page with a P in the thumbnail of the clip that it is lacking frames.
- **Automatically conform missing clips added to Media Pool:** Enabled by default, must be disabled to use collaborative workflow. When this checkbox is turned on, DaVinci Resolve maintains a dynamic relationship between clips in the Media Pool and those in a project’s various timelines. When this checkbox is on and you import clips with matching timecode/file names/reel names to clips in a timeline, DaVinci Resolve will automatically reconform all matching missing clips, and all other timeline clips that have force conform turned off.
- **Assist using reel names from the:** When this checkbox is turned on, DaVinci Resolve uses reel numbers when conforming clips to match any imported project. This setting must also be turned on if you want to choose different reel name extraction methods for individually selected clips using the Clip Attributes window. Turning this checkbox off forces DaVinci Resolve to identify clips using file names when conforming XML and AAF projects. File names can only be used for conforming XML or AAF files, or when importing a DaVinci Resolve project.

There are four options:

Source clip file pathname: Obtains the reel number by extracting it from each media file’s path. This makes it possible to extract a reel number from all or part of the file name, or from all or part of the name of any folder in the path that encloses that file. This extraction is defined using the Pattern field.

Pattern: A code that defines how a reel number should be extracted from the source clip path name.

For more information about creating patterns in “Using the Pattern Field”, see *Chapter 56, “Conforming and Relinking Clips.”*

Media Pool folder name: The reel number is obtained from the name of the bin in the Media Pool that encloses that clip. This option is often used for stereo projects, deriving the reel number from “Left” and “Right” named directories. It’s also useful for projects that are inheriting new VFX clips on a daily basis.

Embedding in source clip file: Useful for file formats where the reel number is embedded within the media file itself. QuickTime files created by Final Cut Pro, DPX frame files, and CinemaDNG files are all formats that are capable of containing reel number header data.

Source clip filename: If there is no defined reel number, often it’s easy to just use the source clip filename. This is a safe option to use in situations where you want to manually choose different reel name extraction methods for individual clips using the Clip Attributes window.

- **Limit reel name matching to X characters:** For situations where you’ve been provided with media files with extra characters in the reel name that don’t correspond to the names used in the project file you’ve been given, “Limit reel name matching to X characters” lets you omit a specific number of characters from the end of a reel name. This works in conjunction with the following setting.
- **Ignore the first X characters of the reel name:** For situations where you’ve been provided with media files with extra characters in the reel name that don’t correspond to the names used in the project file you’ve been given, “Ignore the first X characters of the reel name” lets you omit a specific number of characters from the beginning of the reel name. Combined with the previous setting, you can trim any reel name to a conformable subset of characters.
- **Extract reel names from EDL comments:** Media file formats such as R3D have reel names, obtained from the file names, that are longer than the eight characters that are allowable in a standard EDL. This option allows DaVinci Resolve to extract reel names from appropriately formatted EDL comments, such as those output from Final Cut Pro 7.
- **Sort timeline using reel number and timecode:** Lets you change the behavior of C mode sorting in the Timeline. With this checkbox turned on (the default), all clips in the Timeline are sorted by reel number first, and then by source timecode. This way, clips with similar timecode from the same reel will appear next to one another in C mode. If you turn this checkbox off, reel number is ignored, and all clips in the Timeline are sorted only by source timecode. This may result in clips from multiple sources being mixed together, but it is useful in specific situations.
- **Mixed frame rate format:** (Only available prior to importing media into a project) This drop-down menu lets you choose the method used to conform mixed frame rates for rendering and playback. Which option you choose dictates the accuracy with which retimed clips in DaVinci Resolve match the same clips that were retimed in other editing applications when you import those timelines into DaVinci Resolve via XML or AAF. This drop-down menu also appears in the Load AAF or XML dialogs.

If you’re editing from scratch in DaVinci Resolve: You should leave this setting set to “Resolve.”

When importing timelines via XML from Apple software: Choose the “Final Cut Pro 7” or “Final Cut Pro X” methods of conform.

When importing timelines via XML or AAF from Premiere Pro, Media Composer, Smoke, or other NLEs: You should choose “Resolve.”

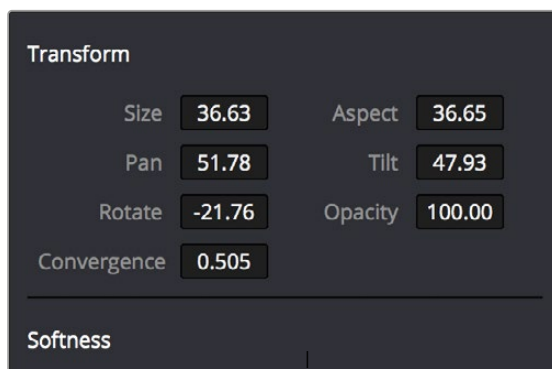
When none is selected: DaVinci Resolve conforms and processes all clips in the Timeline to play at the frame rate that’s selected in the “Timeline frame rate” drop-down menu. For example, 23.98, 29.97, 30, 50, 59.94 and 60 fps clips will all play at 24 fps if that’s what “Timeline frame rate” is set to in the Master Project Settings, and clips will play slower or faster accordingly.

How clips in mixed frame rate timelines are rendered out depends on whether the Render Settings are set to render individual source clips or one single clip. When you render the Timeline as individual source clips, all clips are rendered individually at their original frame rate. If you select “single clip,” all clips are converted to the “Timeline frame rate” frame rate and rendered as a single media file.

Color

These settings affect clip versions and timeline interactions when working in the Color page.

- **Automatically label gallery stills using:** When enabled, DaVinci Resolve automatically generates labels for all gallery stills you take based on the following controls:
 - Naming drop-down:** Lets you choose what name to use for new stills. Options include: Clip Name, Clip Version Name, Source Timecode, Timeline Timecode, Timeline Name, Display LUT Name, Custom Label Using Tags (using metadata variables).
 - Append still number on export checkbox:** When enabled, each new still has an appended still number. Where the number appears depends on the following radio buttons.
 - As Suffix/As Prefix buttons:** Lets you choose to place still numbers at the end of an auto generated gallery label or at the beginning.
- **Luminance mixer defaults to zero:** Selecting this option sets the Y channel of the YRGB parameters for all grades to zero. This is required to be able to export a compliant ASC-CDL, and will impact all grades that use the Lum Mix control.
- **Use legacy Log grading ranges and curve:** DaVinci Resolve 12.5 introduced a modification to the Log grading controls that provides smoother, more pleasing results using the same controls. To maintain backward compatibility with older projects, a “Use legacy Log grading ranges and curve” checkbox in the Color panel of the Project Settings lets you switch your project between the older Log control behavior and the newer one. Older projects that are opened in DaVinci Resolve have this checkbox turned on by default, while new projects have this turned off by default.
- **Use S-curve for contrast:** On by default, this checkbox sets the contrast control in the Color Wheels palette to apply an “S-curve” to the image, such that the shadows and highlights of a signal will not be clipped when you increase the value. If you would prefer for these contrast adjustments to be made linearly, and for the signal to be allowed to clip when you reach the upper and lower boundaries of the video signal, you can turn this checkbox off.
- **Use legacy sizing interactions for windows and effects:** DaVinci Resolve 14.1.1 improved how window tracking applies transformations, to correctly handle things like pixel aspect ratio (par). New projects should leave this setting disabled, however older projects should leave this checkbox enabled to ensure tracking and transforms remain applied the way they were before.
- **Apply stereoscopic convergence to windows and effects:** When enabled, DaVinci Resolve correctly maintains the position of a window that’s been properly placed over each eye as convergence is adjusted in the 3D palette. Enabling this checkbox also enables an additional Convergence parameter in the Window palette that lets you create properly aligned convergence for a window that’s placed onto a stereoscopic 3D clip, as seen in the following screenshot.



The Convergence control in the Transform section of the Window palette appears when you enable “Apply stereoscopic convergence to windows and effects”

- **Use local version for new clips in timeline:** Automatically sets all new clips that are added to existing timelines, or all clips that are added to new timelines that are imported via AAF, EDL, or XML, to use local grades by default. If you want all clips added to new timelines to use remote grades instead, as with DaVinci Resolve version 9 and earlier, you can turn this checkbox off.
- **Automatically match master timeline with media pool:** If you turn on this option before importing any media into the Media Pool, or importing any timelines that will in turn import media into the Media Pool, you can create projects with a Master Timeline. When enabled, clips are added to and removed from the Master Timeline as they're added to and removed from the Media Pool, so that the Master Timeline always contains all media in the Media Pool. Once media has been imported into a project, this setting cannot be changed.
- **Save timeline thumbnails with project:** To minimize project size, and maximize the speed of saving and loading projects, you should leave this checkbox unchecked. If you select the checkbox, all of your Timeline thumbnails will be stored inside every project, instead of in the default directory that's ordinarily dedicated to stills, during both Save and Auto Save operations. This provides a good history of the project but takes much longer to complete and uses more hard disk space.
- **Use BGR pixel order for DPX v2:** Lets you choose a different pixel order for projects using DPX version 2 media.
- **Embed timecode in audio output:** When turned on, directs DaVinci Resolve to output LTC timecode that's embedded in channel 16 of the SDI stream and channel 2 of the analog audio output from your video interface.
- **Use Timelines Bin:** This option is only available to be changed before you add clips to the Media Pool; after you've added clips, it's no longer available. Turning Use Timelines Bin on creates a dedicated Timelines bin in the Media Pool, at the top of the Bin List. When enabled, the Timelines bin contains all timelines in a project, and you're prevented from putting timelines into any other bin in the Media Pool. Whenever you create or import a new timeline, it automatically appears in the Timelines bin. You can add subfolders to the Timelines bin for more specific organization.

Dynamics Profile

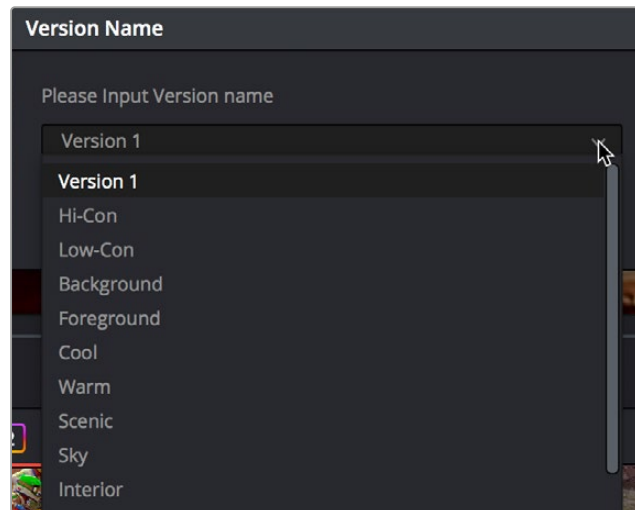
Defines the default transition from one dynamic keyframe to the next for keyframed effects in the Color page. By default this transition is linear, with the "Dynamic profile start" and "Dynamic profile end" parameters set to 1. However, if you need to alter the acceleration of the interpolation of values from one dynamic keyframe to the next, then you can change that keyframe's Dissolve Type in order to "ease" the effect transition you're creating. The values in these settings correspond to the graph curves found in the Dynamic Attributes dialog when editing keyframes in the Color page.

For more information, see *Chapter 148, "Keyframing in the Color Page."* in *"Changing Dynamic Attributes"*.

Versions

Ten text fields provide a way for you to designate automatic names for the versions of grades that you select in the Color page. To the right of each text field, a drop-down menu lets you add a name from a handy list of predefined terms that's been provided. Alternately, you can simply click any field and type your own custom name.

When you change the name of a version in the Color page, the names you define in this list are available from a drop-down menu in the Version Name dialog.



Using the named drop-down when editing the name of a version

Using a predefined list of names for your different versions avoids typos that can later create folder naming issues when you use the “Commercial Workflow” options for rendering your media in the Deliver page.

Camera Raw

This panel contains groups of parameters that correspond to every camera raw media format that's supported by DaVinci Resolve. Using these parameters in the Camera Raw panel, you can override the original camera metadata that was written at the time of recording, and make simultaneous adjustments to all camera raw clips using the “project” raw settings.

To provide more detail, these settings are covered in detail in a dedicated chapter. For more information, see *Chapter 128, “Camera Raw Palette.”*

Capture and Playback

All settings in this panel let you define the functionality of capture and playout to tape using device controlled VTRs connected to your Resolve workstation via the connected video capture and output interface.

For more information on deck capture, see *Chapter 24, “Ingesting From Tape.”*
For more information on video output to tape, see *Chapter 189, “Delivering to Tape.”*

Deck Settings

These settings affect both capture and playback when using the tape ingest options of the Media page, or the tape output options of the Deliver page.

- **Video capture and playback:** You can choose the video format (frame size and frame rate) with which to output to tape from this drop-down menu. HD timelines can be downconverted to SD, and SD timelines can be upconverted to HD using the format conversion of your DeckLink card.
- **Use left and right eye SDI:** A checkbox that enables supported video interfaces to ingest and output muxed stereoscopic video when used with supported VTRs, such as HDCAM SR decks with 4:2:2 x 2 mode. (When muxed stereoscopic signals are ingested, each eye is separated into individual left-eye and right-eye image files.) This parameter only appears when your hardware is set up appropriately.
- **Video connection operates as:** Selects between the available signal options: Use 4:4:4 SDI and Enable Single Link. Which options are available depend on which video capture card you are using.
- **Data Levels:** Lets you specify the data range (normally Video or Full) that’s used when ingesting from or outputting to tape. This option switches the data range of the signal output by your video capture card, but only during capture from tape in the Media page, or output to tape in the Deliver page. When capture or output is not currently occurring, your video capture card goes back to using the identically named data range setting in the Master Project Settings pane, which governs how you monitor the signal being output on an external broadcast display or projector.
- **Video bit depth:** Choose the bit depth that corresponds to the capability of your deck. Depending on your workstation’s configuration, you can choose between 8-bit and 10-bit. Outputting to 10-bit is more processor intensive, but higher quality for compatible devices, and is the default setting.
- **Use deck autoedit:** If supported by your video deck, this is the best method to record video to the deck, as it enables the deck to roll the edit using the specified preroll, and control the edits via serial device control. If this checkbox is turned off, a basic edit On/Off mode is used by the deck, with the potential for frame inaccuracies if the “Non auto edit timing” setting is not properly adjusted.
- **Non auto edit timing:** Adjusts the edit synchronization of the connected deck when auto edit is turned off.
- **Deck preroll:** Sets the number of seconds for preroll. How much is appropriate depends on the performance of your deck.
- **Video output sync source:** When using a DeckLink card this is set to Auto. Other capture cards may require you to set the sync source to “Reference” for playout and “Input” for ingest. This setting is only available if you have a DVS card installed on your system.
- **Add 3:2 pulldown:** Inserts or removes the 3:2 pulldown required to record or play 23.98 fps media to or from a 29.97 tape format.

Capture

These settings are used when you use the Capture mode in the Media page to capture clips from tape into the Media Pool.

- **Capture:** Lets you choose whether to capture both Video and Audio, or Video Only.
- **Video Format:** The format that scanned film frames are saved as. When capturing from tape, the available options are DPX and QuickTime. When capturing from the Cintel film scanner, this is restricted to Cintel Raw Image (CRI), which is a raw data format that DaVinci Resolve automatically debayers as a Cineon log-encoded image for grading.
- **Codec:** The codec used to write captured media. When capturing from tape, these include the various type of Apple ProRes, 8- and 10-bit YUV 422, 10-bit RGB, and the various types of DNxHD. Cintel Raw Image files default to rgb.
- **Save clips to:** A field that displays the directory path to which media files captured from tape are written. You want to choose a volume that's fast enough to accommodate the data rate of the media format you're capturing.

Browse: Click this button to choose a directory to write captured media to. The directory you choose appears in the field above.

- **Save in this folder path:** A series of checkboxes let you specify what other information to use to define the directory hierarchy that will hold the captured media. Every checkbox you turn on adds an additional directory with a name defined by that checkbox's metadata. You can choose any or all of the following: Program name, Clip number, Reel number, and Roll/Card.
- **Apply reel number to:** Lets you choose how to write the reel name. Two checkboxes let you write the reel number to the file's name, and/or to the Header data.
- **Use prefix:** A field lets you type in a prefix to be used in the media file's name. This lets you add text identification that will make the media more easily identifiable and searchable.
- **Apply prefix to:** Two checkboxes let you choose to use the prefix you typed in the file name, and/or in the folder name.
- **Use frame number with:** When capturing to image sequences, you can choose how many digits to use when writing the frame number into the name of each frame file.
- **Set batch ingest handles to:** When capturing to image sequences from a batch list, defines how many frames of additional handles to ingest along with each logged clip.
- **Input:** Lets you choose how many tracks of audio to capture, from 2 to 16.

Playout

These settings only affect the video signal that's output when you use the Edit to Tape mode of the Deliver page.

- **Output:** Lets you choose whether to output both Video and Audio, Video Only, or Audio Only if you're doing an audio layback.
- **Output Source Timecode:** Turn this checkbox on to output each individual clip's source timecode. This option is only applicable when assemble editing to tape.
- **Output LTC:** With a Blackmagic Design DeckLink or UltraStudio device using HD-SDI, longitudinal timecode (LTC) is available on track 16 of the HD-SDI video signal, making it easy to use a Mini Converter de-embedder to extract this analog timecode audio signal and feed it directly to a recording device. This is particularly helpful if you have outboard video processing equipment such as a noise reducer or format converter that passes through the VITC timecode.
- **Delay LTC by x frames:** When outputting LTC to bypass outboard processing gear, such as a noise reducer or format converter, you can compensate for the processing delay by delaying the timecode by a matter of frames to ensure that the processed image and timecode reach the deck at the same time. With a DVS card there is a separate timecode output.

- **Offset audio by x frames:** Lets you specify an offset between the audio track and video to achieve proper A/V sync in cases where the video is being delayed by outboard processing hardware.
- **Output x channels of audio:** Choose the number of audio tracks to output to tape.
- **Set batch payout head handle to x seconds:** When batch outputting multiple clips, you can specify a number of frames before the In point of each clip to be output as well.
- **Set batch payout tail handle to x seconds:** When batch outputting multiple clips, you can specify a number of frames after the Out point of each clip to be output as well.

Subtitles and Transcription

The Subtitles panel lets you adjust presets that govern subtitles being created in subtitle tracks of the timeline.

- **Max Character Per Line:** Defaults to 60. Lets you choose the maximum number of characters allowed on one line in a subtitle.
- **Minimum Caption Duration:** Defaults to 3 seconds. Lets you choose the minimum duration allowed for subtitles in the timeline.
- **Maximum Characters Per Second:** Defaults to 30. Automatically calculates the maximum allowable characters per second based on a subtitle clip's duration.

The Transcription setup panel lets you select the language that DaVinci Resolve uses to analyze audio clips for transcription.

Language: Select one of the supported languages here. Selecting Auto lets DaVinci Resolve analyze the audio clip and choose the language spoken.

Fusion

Fusion allows consistent frame numbering for all compositions, including trimmed clips. This means that any composition created in Fusion can start with the same start frame number. In new projects you can change the default start frame for Fusion compositions in the project settings.

- **Legacy Fusion Composition Frame Count:** Sets the Fusion composition at zero frames to the first frame of the source media for the composition.
- **Default Start Frame:** Lets you type in a default frame number to start your fusion comps in this project.
- **This helps with Visual FX workflows in three ways:**
 - Matching existing keyframes to the right frame offsets—when replacing an asset with mismatched frame offsets (e.g., a sub clip, different format or round trip, or VFX connect workflows with mismatching handles).
 - Rendering to specific frame ranges—when aligning to delivery standards for commercial workflows (e.g., compositions should start at sequence frame 1000).
 - Consistent Referenced Comp behavior—essential when applying a referenced composition to different clips with different trims.

Fairlight

The Fairlight panel lets you set up your project's audio sample rate, as well as setting up various audio-specific tools in the Fairlight page.

Timeline Sample Rate

This setting can only be changed prior to creation of your first timeline. Once one or more timelines have been created in a project, the Audio Sample Rate is locked to whatever was chosen.

The Audio Sample Rate, measured in kilohertz, is the number of samples per second used for audio processing in DaVinci Resolve. This setting defaults to 48000 (or 48 kHz), which is typical for broadcast and cinema work. However, you can change this to 96000 or 192000 if you want to mix and process audio at higher precision. Be aware that using a higher sample rate, such as 96 kHz instead of 48 kHz, will use twice as much processing power and result in media that's twice the size.

NOTE: Regardless of the Timeline Sample Rate you select, when you import audio files at different sample rates, they will be automatically re-sampled to the Timeline Sample Rate so they play correctly.

Bussing

If you want to work using the previous method of fixed-bus mapping, you can do so for new projects by opening the Fairlight panel of the Project Settings and turning on the "Use fixed bus mapping" checkbox.

If your project has fixed busing enabled and you want to change to FlexBus, then uncheck the "Use fixed bus mapping" checkbox. Note that once you have made the change, it will not allow you to change it back to legacy busing. The advantages to changing from legacy busing to FlexBus are enormous, so you will not regret making the change.

Audio Metering

Two options in the General Options of the Project Settings let you customize the Loudness Meters on the Fairlight page, while the others affect all other audio meters in DaVinci Resolve.

- **Meter Type:** Lets you select the desired Meter Type.
- **Level detector:** If the Meter Type is set to Custom, this lets you select Sample Program, VU, or RMS for audio level metering detection.
- **Scale:** If the Meter Type is set to Custom, this lets you select either IEC 60268-18 or Quasi Linear for audio level metering scaling.
- **Decay:** If the Meter Type is set to Custom, this lets you set the time in seconds for level metering decay in 20 dB increments, following peak levels registering on the meters.
- **Peak Indicator:** If the Meter Type is set to Custom, this lets you select the hold time for peak level indication in the meters.
 - Off
 - Short hold + fall
 - Medium hold + fall
 - Medium hold
 - Long hold
 - Infinite hold
- **Pre fader metering on tracks:** Lets you choose how meters in the Fairlight page display their audio analysis. There are two options:

Post Fader (unchecked): Meters always display the level of each clip's signal after whatever fader adjustments have taken place. Fading a track's level down diminishes the visible level of that audio signal in the meter. This setting is good if you prefer a visual indication of the relative levels you've set your various audio tracks to, which is a very NLE-oriented behavior.

Pre Fader (checked): Meters always use the volume levels of the audio clips in that track, even if you've lowered the level using the sliders. If you've keyframed a clip's volume, that change will be reflected by the audio meters, even though fader changes are not. Viewing meters this way means you can always see how much level is available to clips in your mix, regardless of what the current fader levels are set to, in the event you want to keep track of audio you want to bring back into the mix later on. This is a very DAW-oriented behavior.

- **Target Loudness level:** Lets you set the LUFS value that's used as a reference level for loudness metering. Defaults to -23 LUFS, which conveniently makes the display of these meters scale similarly to traditional audio meters that you're already used to.
- **Loudness Scale:** Lets you choose which scale you want to use with which to measure the meters. Options currently include the default of EBU +9 Scale (-18 to $+9$), and EBU +18 Scale (-36 to $+18$).
- **Bus Meter Alignment Level:** Sets the peak of the bus meter.
- **Bus Meter High Level:** Sets the dB level at which the meter starts showing red.
- **Bus Meter Low Level:** Sets the dB level at which the meter starts showing yellow.

Path Mapping

The Path Mapping panel lets you configure your system's file paths, allowing you to seamlessly link and share media clips while collaborating with other users on their own systems. For example, Editor A and Colorist B are collaborating on the same project. Editor A is working on a Mac in L.A., where Colorist B is working on a PC in Bangkok. They are both sharing media in a cloud service's folder, but the file paths to that cloud folder are different for both of them locally.

- The Editor's Mac folder is at `/Users/editor/cloudfolder/Episode 12`
- The Colorist's Windows folder is at `D:\Projects\Episodes\cloudfolder\Episode 12`

Normally if they were collaborating, each one would constantly have to re-link the files from the other before they could continue as the path names would not match. By both of them adding the "Episode 12" location, and mapping their local paths in this section, DaVinci Resolve will automatically convert the file paths on the fly as they work. There would be no need for re-linking the clips as long as all the media they used was in the same hierarchy in the Episode 12 folder.

You can use Path Mapping to make things easier for non-collaborative workflows as well. For example, for a single user on a laptop, they could setup a link from their local media folder to their NAS. This removes the need to relink the media each time they leave and return to the office.

NOTE: Path Mapping differs from using the older Mapped Mount option in the Media Storage preferences. Mapped mount requires each user knowing the file path of all the other users. Path Mapping lets the user just present their own file path, and DaVinci Resolve takes care of the translation for all the other users instead.

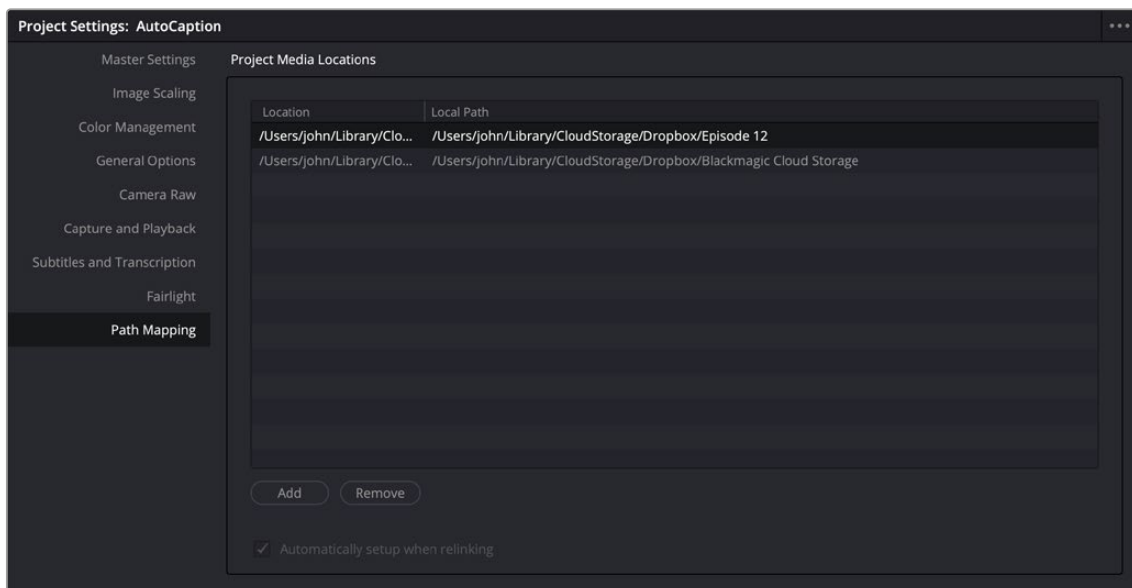
Project Media Locations

This setting lets you set up your project's media locations so that they can easily be shared and translated with other DaVinci Resolve users. The idea is to have a media folder in common with all the other users (i.e., all connected to the same shared cloud storage folder), and set up the path to your own individual folder here.

- **Location:** Shows you the name of the shared folder.
- **Local Path:** Shows the path to this folder on your filesystem.
- **Add:** Opens a filesystem dialog to let you select the shared folder.

Once the location is selected, press the Browse button and use the file system to set the local path to that folder on your computer.

- **Remove:** Removes a shared folder from the Path Mapping settings. It does not delete the folder from your system.
- **Automatically setup when relinking:** With this setting enabled, when relinking offline media, DaVinci Resolve automatically creates a mapping. It compares the path in the project file and the user selected path and creates a relative path between the two. It checks if other offline media in this project is available in those relative paths and relinks them as appropriate. It also does the same check with proxies in proxy subfolders.



The Path Mapping Project Settings